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Unveiling Seven-Decade Disparities of Methane Emissions across China

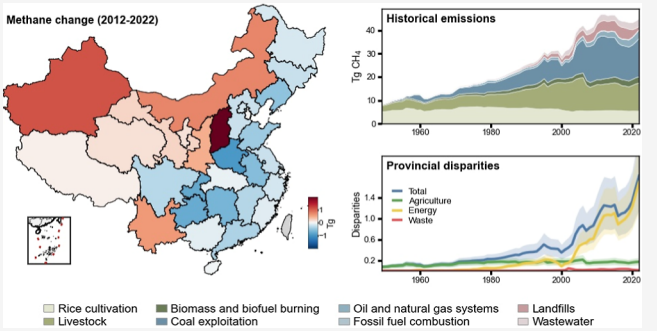
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ABSTRACT: China's century-long economic transition generated substantial methane emissions, yet the trajectory and factors of this evolution remain poorly quantified. A systematic understanding is essential for reconciling China's economic growth with emission mitigation targets and for informing Global South nations. Using a bottom-up estimate of subnational methane emissions (1949–2022), we uncover a fundamental shift from agriculture-dominated to industry-dominated methane emissions in China. Nationwide emissions peaked in 2012, declining 3.8% by 2022, yet provincial disparities widened by 48%. While superficially reflecting heterogeneous fossil fuel use, these disparities are attributable to a shift in explanatory dominance from purely economic to dual economic-policy factors, particularly carbon policies. The long-term stability in agricultural sectors reflects nationwide efforts to food security, which maintains a baseline level of production across provinces. Recent regional decoupling masks the disproportion between emissions and economic gains. Western provinces (e.g., Shanxi, Ningxia, Guizhou) continue to exhibit high-emission but low-return economic profiles. The average of their methane emissions per GDP was 4.8 times that of eastern regions in 2022, with extremes reaching 150-fold differences. For equitable mitigation, future focus should shift from "who emits" to "why emits", establishing a scientific basis for fair compensation between resource-dependent and resource-consuming regions.



KEYWORDS: *anthropogenic methane emissions, emission-economic decoupling, industrial transition, provincial disparities*

1. INTRODUCTION

Methane as a potent short-term climate forcer necessitates urgent mitigation,[1−3] yet its emission profiles are deeply locked in by regions' historical economic trajectories.[4−6] This path dependence, in turn, creates a distributive dilemma for mitigation policies, and one-size-fits-all approach risks imposing disproportionate burdens on vulnerable regions with limited historical responsibility, potentially exacerbating challenges such as energy poverty, agricultural disruption, and fiscal instability.[7−9] Given the short-lived nature of methane, the principle of common but differentiated responsibilities calls for a critical examination of the divergent developmental pathways creating regional high-emission systems.[10−12] This perspective shifts the focus from merely identifying current emitters to understanding the underlying factors (e.g., economic structure, energy systems, and technological trajectories).[3,12,13] Therefore, effective and just mitigation requires addressing these determinants, while balancing methane reduction against food security, livelihoods, and equitable development.[14]

China's long-term economic transition provides a valuable case for examining the coevolution of development and methane emissions, yet a systematic quantification of this synergy is lacking. The socio-technical systems underpinning methane emissions (e.g., energy infrastructure, land-use patterns, and agricultural practices) exhibit strong path dependence and institutional inertia.[3,15,16] A long-term analysis can capture the complex interplay among developmental stages, policy choices, and emissions, thus illuminating responsibility embedded in both historical and current outputs.[17,18] Such insights are vital not only for achieving China's dual carbon goals but also for informing other Global South nations pursuing low-carbon development.[3,5] Existing studies estimating methane emissions via bottom-up or top-down approaches often focus on isolated years or short time spans.[11,19−21] These estimates, particularly from bottom-up methods, show high uncertainty, largely because emission factors (EF) are sensitive to local conditions and activity data are often inconsistent across sectors.[17,22] Such issues hinder both regional comparability and temporal consistency. Although remote sensing data provide valuable spatial

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coverage, their temporal scope is limited to the past two decades.[20,23,24] These limitations constrain a robust analysis of spatiotemporal dynamics in methane mitigation. Establishing long-term subnational inventories is therefore a foundation for designing sector-specific and differentiated policies and addressing responsibility-benefit mismatches.

Provincial prioritization of sector-specific methane mitigation strategies requires a clear understanding of the key factors associated with regional disparities,[1,11] especially the roles of economic activities and policy implementation in changing methane emissions.[1] However, its long-term dynamics and the linkage to socioeconomic transition remain underexplored, that existing studies tend to be fragmented, focusing either on isolated sectors or aggregated emissions, failing to provide an integrated and multisectoral perspective.[3,4,25,26] Given opposing economic and policy pressures across sectors, a fragmented perspective risks sectoral mitigation gains being offset by emissions growth elsewhere, challenging overall climate governance.[3,27,28] For China, detailed knowledge of the factors underlying long-term methane disparities is essential to designing transition-sensitive policies amidst rapid structural change.

This study addresses the under-quantified historical dynamics of methane, regional disparities, and their factors under China's economic transition. Applying IPCC guidelines and localized accounting parameters, we construct a comprehensive subnational emission inventory for Mainland China spanning 1949−2022. It enables the evaluation of annual emission trends within agriculture, energy, and waste sectors. By quantifying the contribution of key provinces and sectors to regional disparities via semivariogram analysis, we reveal the shift of dominant sectors and identify stage-specific impacts of economic-environmental policy factors. Specific research questions include: (1) what characterizes the spatiotemporal heterogeneity of methane emissions through China's developmental transitions? (2) Which province-sector combinations disproportionately influence regional disparities of methane emissions, and (3) what role do economic activities and environmental regulations play in affecting these dynamics?

2. MATERIALS AND METHODS

2.1. Methodology for Subnational Methane Inventory

Subnational anthropogenic methane emissions in China were estimated from three dominant sectors: agriculture (including livestock, rice cultivation, biomass and biofuel burning), energy (including coal exploitation, oil and natural gas systems, and fossil fuel combustion), and waste management (including landfills and wastewater management). The methods outlined in the 2006 IPCC Guidelines for National Greenhouse Gas and the 2019 Refinement[29,30] are used to estimate methane emissions for these eight sectors together with localized EF

$$\text{methane}(S, t) = \sum_{C} \text{AD}_{S,C}(t) \times \text{EF}_{S,C}(t) \times (1 - \text{CF}_{S,C}(t)) \quad (1)$$

where methane(S,t) represents the total methane emissions from sector S in the inventory year t. The parameter $\text{AD}_{S,C}(t)$ refers to the activity data in the year t for sector S and condition C. CF is the correction factor, which indicates the fraction of methane utilized or oxidized without being released to the atmosphere.

2.1.1. Activity Data, EFs, and Uncertainty of Agriculture Sectors. Methane emissions from rice cultivation were estimated using the methodology,[17,31] that compared different EFs for rice cultivation in China and provided the EFs of each rice season by region suitable to the Chinese situation (Table S3). Province-level annual rice cultivation areas (early rice, middle rice, and late rice) were collected from China Agricultural Statistical Yearbook and China Rural Statistical Yearbook. The growing days for early, middle, and late rice are 77, 110−130, and 93 days, respectively.[17]

Methane emissions from livestock were estimated as the sum of methane emissions from enteric fermentation and manure management. Province-level annual census data of domestic livestock for each livestock category were collected from China Agricultural Statistical Yearbook, China Rural Statistical Yearbook and New China 60 Years Statistics Compilation. Because seasonal births and slaughters change the population of livestock, we used the slaughtered population and live population at the end of the year to estimate the total emissions. The average life spans are 12 months for dairy cattle, 10 months for nondairy cattle and buffalo, 7 months for sheep and goats, 6 months for swine, and 2 months for poultry. For their EF, we used the values from ref 32. The maximum and minimum values for each category were applied to the uncertainty analysis.

Methane emissions from biomass and biofuel burning come from the straw in rural households and the burning of firewood. The former mainly refers to crop residues used as biofuels in houses and burnt in open fields. The total crop residues are calculated as annual crop yields and straw/grain ratio for the crop: rice (0.623), wheat (1.366), corn (2), other grains (1), bean (1.5), tuber (0.5), cotton (3), fiber crop (1.7), sugar crop (0.1), and oil-bearing crop (2).[31] We collected the province-level production data from China Agricultural Statistical Yearbook and used aburning efficiency 100% for biofuels and 88.9% for fires in open fields.[17] For the estimation of methane emissions from firewood burning, we collected the province-level data from China Energy Statistical Yearbook, with the EF_s of crop residues for biomass fuel 3.62 ± 2.20 kg CH_4 t^{-1} and fire in open fields 3.89 ± 2.20 kg CH_4 t^{-1}, respectively. For the firewood consumption, the EF_s 2.77 ± 1.80 kg CH_4 t^{-1} was adopted.[17]

2.1.2. Activity Data, EFs, and Uncertainty of Energy Sectors. Methane emissions from coal exploitation include fugitive methane from underground, surface, and postmining activities. We collected the province-level coal production data from China Energy Statistical Yearbook and China Statistical Yearbook. For methane emissions from underground mines, we used the mean EF_s that obtained from 787 coal mines across 25 provinces for the period 2005−2010 (Table S3).[33] The minimum and maximum values were applied to analyze uncertainties. Methane emissions from surface mining are generally much lower than those from underground mining. On average, only about 5% of coal was mined from surface mines at the national scale.[33] Due to the limited availability of methane emission measurements for surface mining, we adopted the default EF of 2.5 m^3 ton^{-1} of coal mined from IPCC.[29] We assumed that 12% of the total emissions from coal mining are associated with emissions during the subsequent handling, processing, and transportation of coal.[33] In China, the fraction of methane collected and utilized from coal mining operations ($\text{CF}_{S,C}$) has increased with economic growth and improvements in coal safety. Based on the findings,[17,33] the recovery fraction increased linearly from 3.59% in 1994 to 9.4% in 2010. We used a recovery fraction of 3.59% for the period before 1994 and 9.4% after 2010. To account for uncertainty, we adopted a range of recovery fractions (3.59%−5.21%).[17,33]

Methane emissions from oil and natural gas systems were treated as an aggregated source under IPCC subcategory 1B2 (Fugitive Emissions from Oil and Gas), adopting the methodologies from refs 17 and 34. We collected the province-level annual crude oil and natural gas production data from the China Statistical Yearbook. For oil systems, we applied the EF_s of 0.077 kt CH_4 PJ^{-1}, with an uncertainty range of 0.058−0.190 kt CH_4 PJ^{-1}. For natural gas systems, we used the fugitive emission rates (FERs), which assume a linear decrease in FER from 4.6% (equivalent to 0.81 kt CH_4 PJ^{-1}) in 1980 to 2.0% (0.35 kt CH_4 PJ^{-1}) in 2010.[17] This aligns with a FER of 1.9% reported for the Organisation for Economic Co-operation and Development (OECD) countries. Uncertainty was assessed using two scenarios: (1) a lower FER scenario, where the FER in China

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Environ. Sci. Technol. XXXX, XXX, XXX−XXX

decreased from 3.2% in 1990 to 1.4% in 2020; and (2) a higher FER scenario, where the FER decreased from 5.4% in 1990 to 4.6% in 2020.

Province-level fossil fuel combustion data were collected from China Energy Statistical Yearbook. We adopted EFs of 1 kg CH_4 TJ^{-1} for coal combustion, 3 kg CH_4 TJ^{-1} for oil combustion, and 1 kg CH_4 TJ^{-1} for natural gas combustion, as recommended by IPCC.[29,30] The uncertainty is set 60% of the EFs for this sector.[17]

2.1.3. Activity Data, EFs, and Uncertainty of Waste Management. Methane emissions from landfills were estimated using the first-order decay method according to IPCC guidelines

$$\text{methane}(t) = (1 - e^{-k}) \times \sum_{x} e^{-k\times(T_L - x)} \times \text{MSW}_L \times \text{MCF}_T \times F_T \times \text{DOC} \times \text{DOC}_d \times f \times (1 - O_f) \times 16/12 \quad (2)$$

where k is the reaction constant (0.3) and T_L is the decay lifetime period (4.6 years).[17] MSW_L is the total amount of municipal solid waste treated by landfills. MCF_T represents the correction factor from three types of landfills: 1.0 for managed anaerobic landfills (Type_A), 0.8 for deep unmanaged landfills exceeding 5 m (Type_B), and 0.4 for deep unmanaged landfills within 5 m (Type_C).[29,30] F_T is the fraction of MSW_L assigned to each landfill type by province, as shown in Table S4. DOC is the fraction of degradable organic carbon in MSW, DOC_d is the fraction of DOC that can be decomposed, f is the fraction of methane in landfill gas, and O_f is the oxidation factor. We adopted a DOC of 6.5%, DOC_d of 0.6, f of 0.5, and O_f of 0.1.[17,32] For uncertainty analysis, we assumed maximum methane emissions with DOC_d = 0.6 and f = 0.6, and the minimum emissions with DOC_d = 0.5 and f = 0.4. The province-level MSW treated by landfills was collected from China Environmental Statistical Yearbook and China Statistical Yearbook.

Methane emissions from wastewater management arise from industrial wastewater and domestic sewage. We collected the data from China Environmental Statistical Yearbook and China Statistical Yearbook and estimated the emissions using eq 3

$$\text{methane}(t) = \text{COD}(t) \times \text{EF} \times \text{MCF} \quad (3)$$

The degradable organic fraction is the main factor for industrial wastewater, commonly represented by chemical oxygen demand (COD). We used the default EF_s 0.25 $gCH_4/gCOD$ for the COD removed. The correction factor (MCF) for the COD removed through wastewater treatment plants (WTPs), and that for the COD discharged into waterbodies is assumed to be 0.458 and 0.1.[17,29,30] The emissions from sewage management are usually expressed using biochemical oxygen demand (BOD_5). We used the COD content in domestic sewage and regional BOD_5/COD ratios to estimate BOD_5 since the provincial values cannot be obtained directly from statistical data (Table S4). The MCF for BOD_5 removed through WTPs and that for BOD_5 discharged into waterbodies was set at 0.165 and 0.1.[17,29,30] For uncertainty, we used MCFs of 0.3 (domestic sewage) and 0.5 (industrial wastewater) for maximum emissions while MCFs of 0.1 (domestic sewage) and 0.2 (industrial wastewater) for minimum emissions.[29,30]

2.2. Methods to Handle with Missing Data for the Period 1949−2022

The first China Statistical Yearbook was released in 1982, and the data gaps for the prereform and opening-up period need interpolation and extrapolation. This study adopts the methodology from refs 17 and 18. First, the activity data are anchored at the earliest ($t1$) and most recent years ($t2$) with available statistical data for each province and sector. We employed linear interpolation to estimate missing values between these anchor years based on historical trends. For periods preceding $t1$ and following $t2$, national-level statistics are prioritized, and then provincial data are allocated according to their average proportion to the national total over a nearest multiyear reference window adjacent to $t1$ or $t2$ (typically four or five consecutive years, and broadly consistent with Five-Year Plan reporting periods where applicable). Given the relative stability of economic development policies during the prereform planned economy era, this allocation method is reasonable when detailed historical provincial data are unavailable. Indeed, 60 Years of New China Statistical Data Compilation provides national and provincial-level data for most sectors before 1980.

However, data gaps still persist, mainly waste sectors. Considering the high correlation between waste management and population as well as economic data,[17,18] we employed linear association to estimate the national activity data of landfill and COD in wastewater discharge. Additionally, administrative reallocations are applied where territorial boundaries changed (e.g., Hainan province merged into Guangdong before 1988; Chongqing separated from Sichuan after 1996). To quantify activity-data uncertainty, relative mean errors of the interpolation and extrapolation procedure were estimated by leave-one-out temporal cross-validation for each province and sector, and a two times mean-error bound was applied to filled years within the corresponding gap-filling windows. We further assessed sensitivity by applying province- and sector-specific ±10−20% activity perturbations and propagating them through the emission inventory. Figure S1 shows the availability of activity data that can be obtained from multiple Statistical Yearbooks directly.

2.2.1. Agriculture Sector. For rice cultivation, where subtype area data were incomplete in the early years, each subtype's share of the total rice area was extrapolated using a quadratic function of national GDP. These shares were then multiplied by the total national rice area to reconstruct the provincial areas. Provincial subtype areas before $t1$ are allocated from the corresponding national series using each province's mean share of the national subtype total over a short reference window after $t1$, as appropriate for each subtype. Remaining provincial gaps are filled by linear interpolation between anchor years.

For livestock, the following procedures were applied. At the national scale, sheep and goat inventories were linearly interpolated, and their slaughter numbers derived from inventory−slaughter ratios fitted against time. Cattle subpopulations before $t1$ were split from total cattle using mean subtype shares over an early reference window. Poultry live and slaughter numbers were linked to GDP through linear ratio regressions, and early gaps in swine slaughter data were extrapolated by linear regression. Provincial livestock data before $t1$ are allocated from the harmonized national series using provincial shares over the reference window. Provincial series are then linearly interpolated between anchor years.

For biomass and biofuel burning, provincial crop production before $t1$ is allocated from national crop totals using mean provincial shares over the reference window after $t1$, and the other grains are treated using provincial mean absolute levels over the same window where appropriate. For fuelwood consumption, national gaps are estimated by linear association with national population, and then provincial fuelwood in periods with sparse observations is allocated from the national total, followed by linear interpolation of remaining provincial gaps.

2.2.2. Energy Sector. For coal, oil, and natural-gas production and fossil-energy consumption, the national-level production and consumption series were adopted as the anchor values. Provincial values before $t1$ are allocated from national totals using each province's mean share over the reference window.

2.2.3. Waste Management Sector. For landfilled waste, when early national data were unavailable, waste generation was reconstructed as follows: the waste-to-GDP ratio was regressed linearly against national population, and the predicted ratio was subsequently multiplied by national GDP. Provincial landfilled waste before $t1$ is allocated from the national total using mean provincial shares over the reference window after $t1$.

For wastewater, internal national data gaps were addressed by linear interpolation against GDP. Early national totals are further estimated by linear association with GDP. Industrial discharged/disposed splits and domestic discharge shares are reconstructed from observed ratios or fixed mean shares where records are incomplete. Missing national COD is estimated from observed COD-to-wastewater ratios in later statistical years, with GDP-based linear

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Environ. Sci. Technol. XXXX, XXX, XXX−XXX

extrapolation where needed. Provincial COD for each discharge/disposal pathway is allocated from national COD totals using mean provincial shares over sector-specific reference windows adjacent to $t1$, and provincial series are linearly interpolated against GDP where internal gaps remain before allocation.

2.3. Semivariogram Analysis for Disparities Estimation of Methane Emissions

The provincial disparities were quantified using a binned semivariance approach. This approach characterizes the variation in emission disparities as a function of geographic separation distance.[35−37] Annually, we computed the absolute pairwise differences in methane emissions between all unique pairs of provinces, using the geographic distance ($d_{i,j}$) for each province pair (i,j). These pairwise differences were then grouped into predefined distance bins defined by distance bin edges (bin_edges). Specifically, a pair (i,j) was assigned to appropriate bin k if $d_{i,j} \in [bin_edges_k, bin_edges_{k+1})$. The semivariance for bin k was then computed by eq 4

$$\overline{\gamma}_k = \frac{1}{2N_k} \sum_{(i,j)\in k} (\mathrm{CH4}_i - \mathrm{CH4}_j)^2 \tag{4}$$

where $\overline{\gamma}_k$ is the average differences of methane at the spatial scale defined by bin k, and N_k is the number of province pairs. The overall disparities metric I was then derived as the mean semivariance across all nonempty bins

$$I = \frac{1}{K} \sum_{k=1}^{K} \overline{\gamma}_k \tag{5}$$

where K is the number of valid bins. Empty bins (no province pairs within the specified distance range) were excluded from the final calculation to avoid bias, and a higher I value indicates greater regional disparities.

Gini coefficient and Theil index are more common in calculating the disparities,[3,38] but it only measures overall distribution disparity. However, this study targets spatial disparity in provincial methane emissions across China. Methane is released from geographically localized sources, such as rice cultivation, coal mining, and oil and gas systems, so provincial differences carry both geographic structure and absolute magnitude,[1,5,39] with direct implications for where national mitigation efforts should be concentrated.[5,6] Specially, Gini and Theil indices summarize only relative shares among provinces and can be misleading for assessment: (1) they are location-blind and cannot represent how emissions cluster across the landscape; (2) they may mask widening absolute gaps, and thus rising regional abatement burdens when shares converge during rapid emission growth; (3) they can imply easing disparity when relative shares become more but spatial heterogeneity in absolute emissions still intensifies, potentially misdirecting regional reduction priorities (Figure S2). Semivariance explicitly captures spatial dependence through distance bins (e.g., 0−500 km, 500−1000 km), reflecting both spatial autocorrelation and amplified cross-regional differences via squared differences. Figure S3 shows the sensitivity of semivariance to different distance bin selections.

2.4. Contribution of Explanatory Factors to Provincial Disparities of Methane Emissions

This analysis spans the period 1995−2022, constrained by data availability of factors. This temporal scope concurrently covers the observed increase in the disparities. 65 economic and environmental indicators were selected, categorized into eight groups (Table S1): industrial activity factors, agricultural activity factors, energy structure factors, economic and urban-rural differences factors, waste management factors, environmental governance investment, carbon-related policies, and agricultural and ecological policies. All indicators (except methane-related policies) were obtained from the National Bureau of Statistics of China. The indicators belonging to methane-related policies were sourced from Subnational Methane Monitor (https://subnationalch4.com/). The first four categories represent economic structure and development dimensions, whereas the latter three pertain to environmental governance and investment. To mitigate multicollinearity concerns, we conducted annual principal component analysis within each category, retaining components that cumulatively explained ≥85% of the variance. For each year, we generated a Monte Carlo sample of spatial disparity observations that we randomly drew 20 of the 31 provinces without replacement, computed the spatial disparities of provincial methane emissions as the response, and computed the spatial disparities of the retained principal components within each of the eight indicator categories as predictors. This procedure yields 50,000 within-year samples with eight category-level disparity predictors (Figure S4).

We employed XGBoost-SHAP framework to assess the contributions of the above indicators to observed disparities. Bayesian optimization was applied to enhance predictive accuracy while reducing hyperparameter tuning costs.[40,41] XGBoost, a gradient-boosted decision tree algorithm, leverages first and second-order derivatives combined with regularization terms in the objective function (Obj). This dual approach prevents overfitting by simultaneously minimizing prediction error from loss function and constraining model complexity.[40,41]

$$\mathrm{Obj} = \sum_{i=1}^{N} L(I_i, \hat{I}_i) + \Omega \sum_{k=1}^{K} (f_k) \tag{6}$$

$$\Omega(f_k) = \gamma T + \frac{1}{2}\lambda \|\omega\|^2 \tag{7}$$

where N is the number of training samples, $L(I_i, \hat{I}_i)$ is the loss function characterizing the difference between the actual and fitted values, $\Omega(f_k)$ is the regularization term used to control model's complexity, T is the number of leaves in tree K; ω is the leaf weights of target variable on leaves; γ is the leaf penalty coefficient controlling tree growth, and λ is the regularization term constraining leaf weights.

Shapley Additive exPlanations (SHAP) can interpret the output results of XGBoost models and measure the contribution of a predicted value to each feature based on marginal contribution.[41] Using XGBoost to predict the disparities, we applied SHAP to allocate the contribution of eight key variable categories, as represented by eq 8

$$\varphi_i = \sum_{S\in N} \frac{|S|!(n-|S|-1)!}{n!} [f_{S\cup\{i\}}(x_{S\cup\{i\}}) - f_S(x_S)] \tag{8}$$

where φ_i is the contribution of feature i, N is the group with n features, S is the full subset of features without feature i, $|S|!$ is the factorial of the number of features contained in S, x_S is the input value in S, $f_{S\cup\{i\}}$ is the model trained with feature i, and f_S is the model trained without feature i. This same methodology was extended to identify key provinces and sectors contributing to the disparities of methane emissions. By excluding methane emissions from individual provinces or sectors and quantifying the resulting changes in disparities, we determined their relative impacts. For the detailed workflow, please see Figure S4.

2.5. Calculation of Decoupling Index and Disproportion Index

The decoupling theory posits a critical disjunction between economic growth and its environmental footprint.[42] It contends that advanced stages of development can and should achieve a dissociation where economic expansion no longer necessitates increased resource consumption or exacerbates environmental pollution. This theory advocates for a transition toward absolute decoupling, where economic growth proceeds concurrently with an absolute decline in resource use and environmental degradation, thereby positioning itself as an essential theoretical foundation for sustainable development and the transition to a green economy. We used the Tapio decoupling model,[43] as follows

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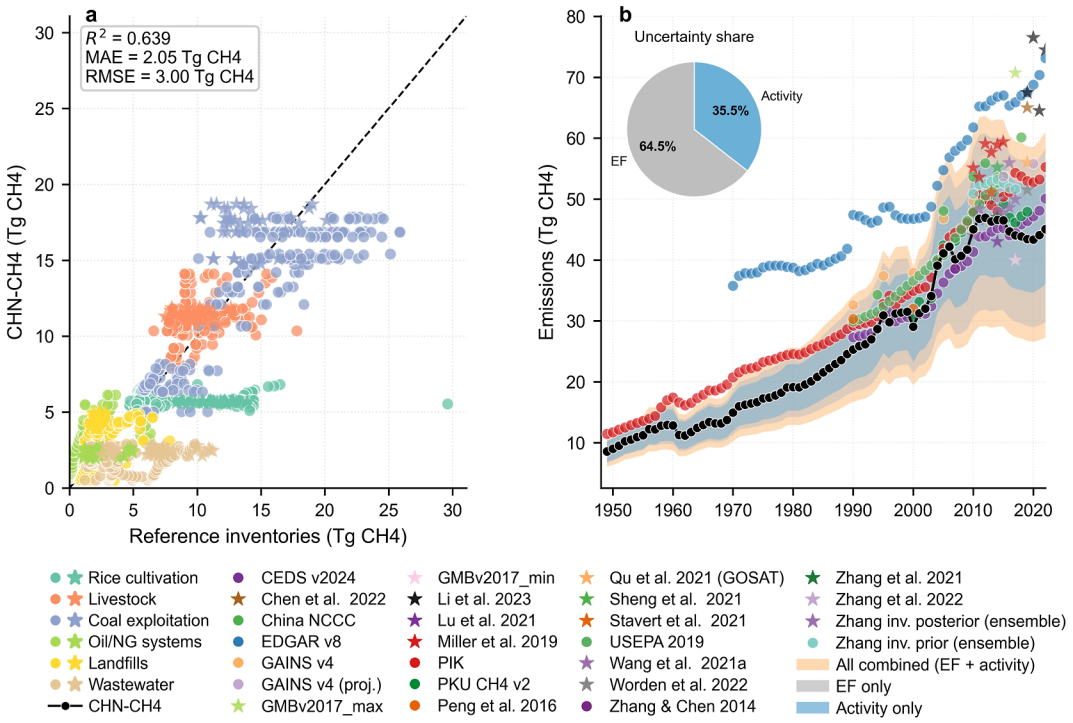


Figure 1. Comparison of estimated anthropogenic methane emissions (CHN−CH4) with the references and uncertainty attribution for national totals (1949−2022). (a) Sectoral CHN−CH4 estimates versus reference inventories. Circles denote bottom-up inventories, and stars denote top-down estimates, while colors indicate emission sectors. (b) National total anthropogenic methane emissions. The black line shows the CHN−CH4 central estimate. Shaded bands indicate activity-only uncertainty, EF-only uncertainty, and the combined from both sources. The inset pie chart gives the mean fractional contribution of EF versus activity uncertainty to the national emission uncertainty.

$$t(\Delta CH4, \Delta GDP) = \frac{\Delta CH4}{\Delta GDP} = \frac{(CH4_{year} - CH4_{year-1})/CH4_{year-1}}{(GDP_{year} - GDP_{year-1})/GDP_{year-1}} \quad (9)$$

In eq 9, $t(\Delta CH4,\Delta GDP)$ denotes the decoupling status between methane emissions and the value added in the corresponding sectors. The decoupling status is classified into eight categories,[43] including (1) expansive negative decoupling ($t > 1.2$, $\Delta CH4 > 0$, $\Delta GDP > 0$); (2) weak negative decoupling ($0.8 > t > 0$, $\Delta CH4 < 0$, $\Delta GDP < 0$); (3) strong negative decoupling ($t < 0$, $\Delta CH4 > 0$, $\Delta GDP < 0$); (4) recessive decoupling ($t > 1.2$, $\Delta CH4 < 0$, $\Delta GDP < 0$); (5) weak decoupling ($0.8 > t > 0$, $\Delta CH4 > 0$, $\Delta GDP > 0$); (6) strong decoupling ($t < 0$, $\Delta CH4 < 0$, $\Delta GDP > 0$); (7) expansive coupling ($1.2 > t > 0.8$, $\Delta CH4 > 0$, $\Delta GDP > 0$); and (8) recessive coupling ($1.2 > t > 0.8$, $\Delta CH4 < 0$, $\Delta GDP < 0$).

We also quantified the disproportionality index between methane emissions and economic output, by calculating the ratio of a province's share of national methane emissions to its share of national GDP for each year. Values above 1 indicate higher emissions relative to economic contribution, whereas values below 1 reflect lower emissions alongside higher economic output. This approach is consistent with methods used in regional environmental-economic studies.[44]

3. RESULTS

3.1. Validation of Subnational Anthropogenic Methane Emissions

We compared estimated methane emissions (CHN−CH4) with reference inventories at sectoral, regional, and national levels (1949−2022), ensuring comparability and comprehensiveness. Figure 1 benchmarks CHN−CH4 against 31 bottom-up (BU) inventories and 15 top-down (TD) inversions. Figure 1a uses all available sector-level BU pairs (n = 1002), yielding R^2 = 0.64, MAE = 2.0 Tg, and RMSE = 3.0 Tg. TD sector inversions (n = 240) show similar scatter (MAE = 3.1 Tg). Figure 1b extends the national total over time. 65% of independent totals with uncertainty bounds fall within the CHN−CH4 envelope (79% for 1990−2020, where sector benchmarks overlap). Averaged over 1949−2022, emission-factor uncertainty accounts for most of the national uncertainty (64.5%), with activity-data uncertainty contributing the remaining 34.5%. The long-term PIK series correlates strongly with CHN−CH4 (r = 0.99), supporting the credibility of the historical reconstruction.

Provincial totals agree closely with PKU-CH4_v2 (r = 0.90, MAE = 0.30 Tg, n = 580) and correlate positively with EDGAR v8 (r = 0.62, n = 899), with robust regional consistency (Figure 2). EDGAR v8 is systematically higher than CHN−CH4 (49% mean total) and 34% above the median of peer BU inventories. This is dominated by rice cultivation (2.4 times CHN−CH4), likely reflecting higher rice EF and continuously flooded areas in EDGAR, with secondary inflation in wastewater.

Regarding the post-2010 emission plateau, CHN−CH4 peaks in 2012, with 8/9 benchmark inventories at that peak year inside the CHN−CH4 uncertainty range. Peak-year analysis (2005−2020) shows broad consensus on an early-2010s maximum: 13 of 26 national trajectories (50%) peak in 2010−2015, including 6 of 12 BU estimates (50%) and 7 of 14 TD inversions (50%). CHN−CH4 lies within this consensus band. Together, these patterns indicate that CHN−CH4 credibly represents the timing and magnitude of China's national methane emission peak.

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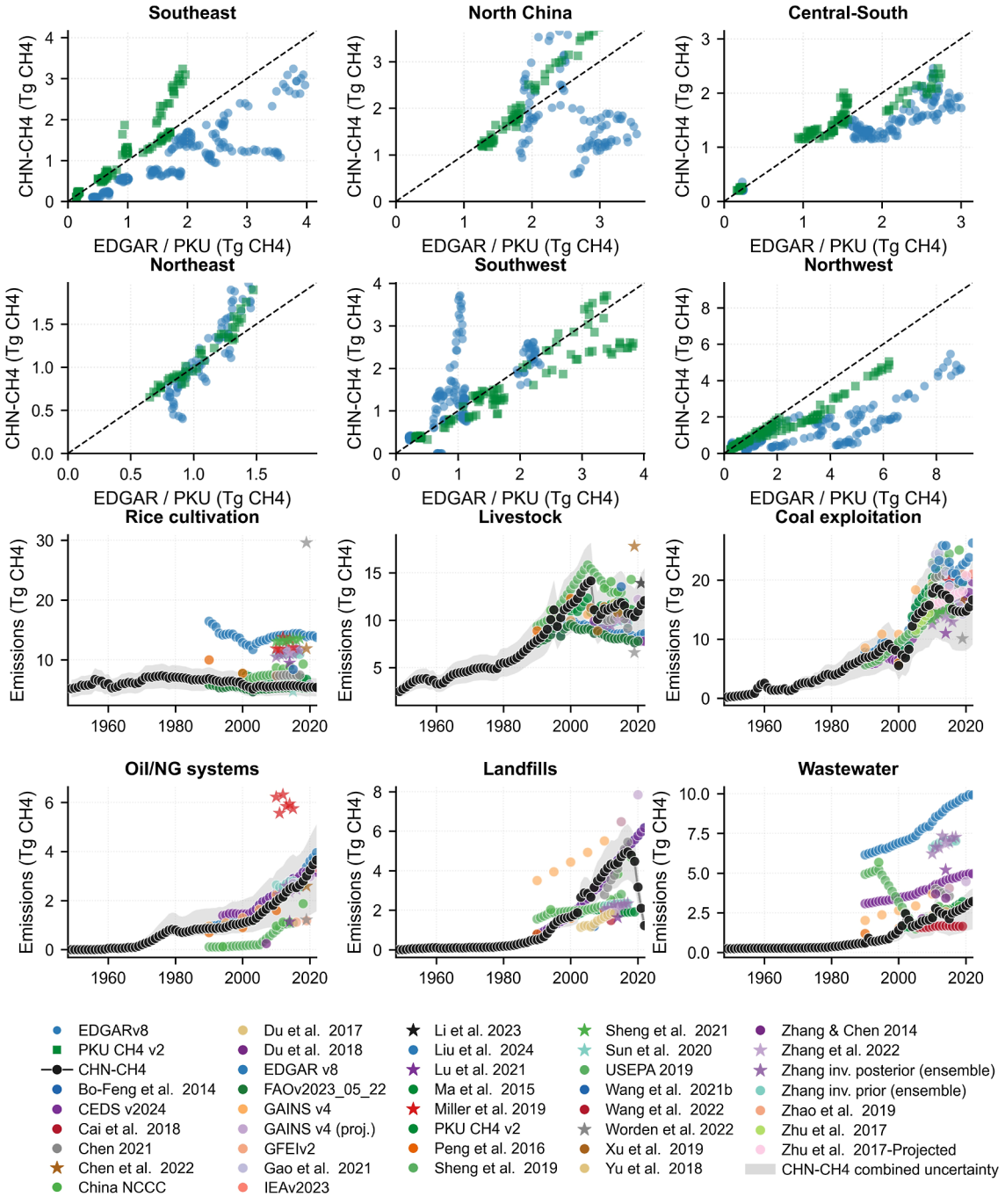


Figure 2. Comparison of CHN−CH4 with EDGAR v8 and PKU-CH4_v2 across six Mainland China regions, including sectoral disaggregation against reference inventories.

3.2. Polarized Trends of Methane Emissions across China's Provinces

China's anthropogenic methane emissions experience a transition in dominant sectors from agriculture to energy, while waste generation rises substantially after 2000 (Figure 3a). The long-term rise in national emissions closely aligns with major policy shifts: post-1949 industrialization, post-1978 economic reforms, and post-2000 market liberalization and globalization (Figure 3a). The period with the highest magnitude and rate of increase was 2000−2010, during which the decadal increase surpassed the accumulation increase of the preceding three decades. Per capita metrics offer a distinct perspective. In 2022, China's per capita emissions stood at 0.03 tCH_4, remaining below the global average of 0.05 tCH_4, and the southern provinces had retreated to their pre-economic reform (post-1978) levels (Figure 3b). These achievements benefit from the intervention of methane mitigation policies, including both direct and indirect measures (e.g., Air Pollution Prevention and Control Action Plan, Methane Emission Control Action Plan).

Provincial methane emissions show high spatiotemporal heterogeneity (Figure 4). The southeastern provinces (e.g., Guangdong, Guangxi, and Jiangxi) exhibited the highest emissions, primarily due to rice cultivation before 1978 economic reforms. Following rapid urbanization after 2000, there was a marked shift away from agriculture (decreasing

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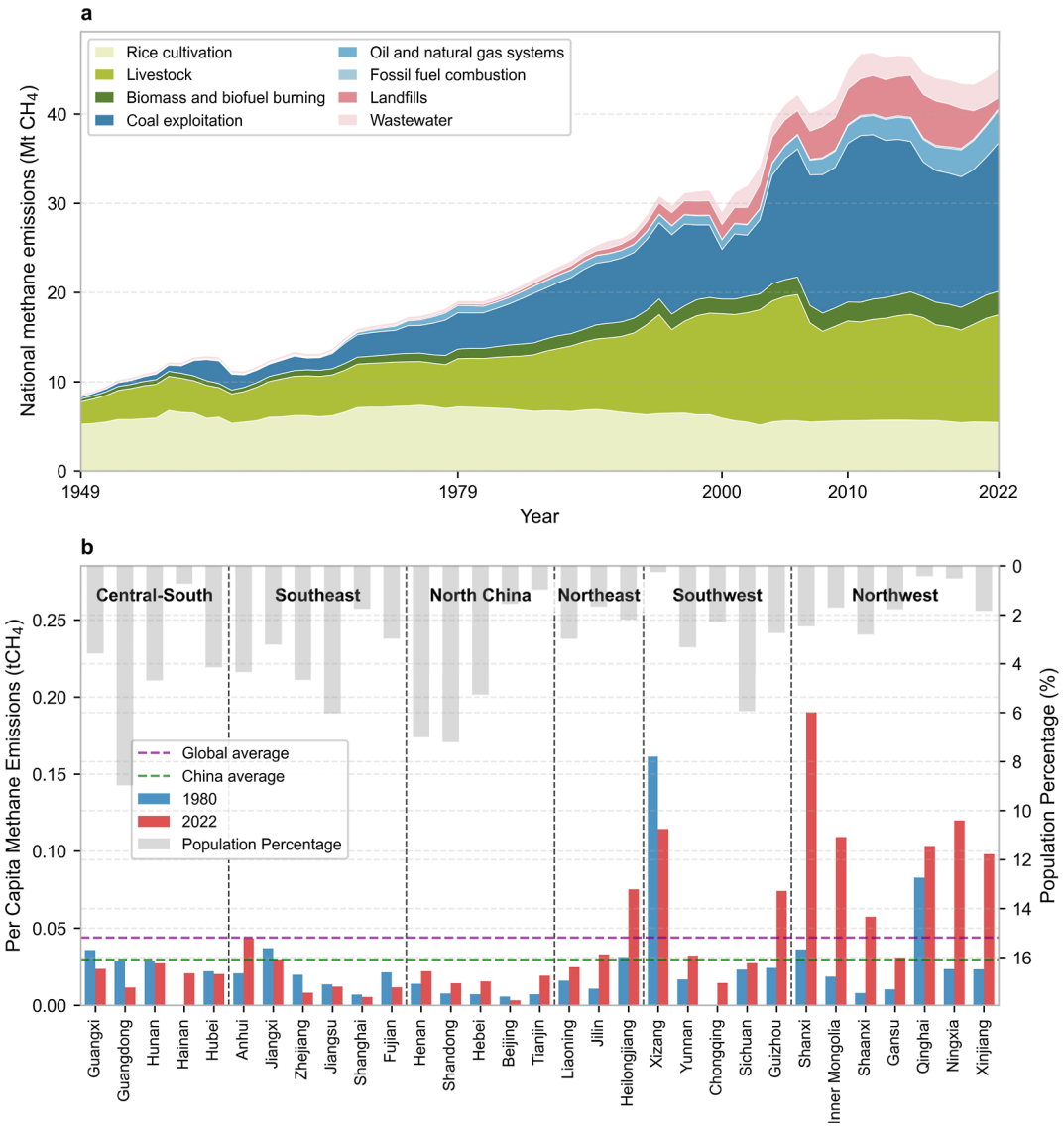


Figure 3. Historical changes of anthropogenic methane emissions in Mainland China from 1949 to 2022. (a) National and sectoral changes of methane emissions. (b) Differences in per capita emissions and population share across provinces between 1980 and 2022. The global average of per capita emissions in 2022 is sourced from the World Bank.

from 0.83 in 2000 to 0.71 Tg in 2022), toward a more resource-intensive industrial and urban development model.[45,46] Zhejiang and Shanghai experienced the most reductions, decreasing by 58% and 51% over this period, respectively. However, this shift may have transferred a growing burden of resource consumption and emissions to northern China, notably the northeastern provinces, where the average rice paddy area expanded by more than 63% (Table S2). The northern and western regions saw an almost 3.5 times surge in average emissions, with 4.47 times from energy sectors between 1980 and 2022. Prior to 2000, western regions were dominated by livestock-related emissions. After 2000, a shift toward energy-related sectors drove substantial emission growth, particularly after 2005 (Figure 4). Shanxi's coal mining sector exemplifies this, with emissions increasing 3.8-fold between 2000 and 2010.

The divergent pattern of methane emissions features polarization post-2010 (Figures 4 and 5a,b). The northwest region shows sustained growth, primarily associated with coal-related energy activities, while most developed eastern provinces exhibit declining trend (Figures 4 and 5b). Even within the coal mining sector, central and eastern provinces such as Henan, Shandong, and Anhui have shown substantial reductions. This turn stems from China's consolidation of the coal mining sector, which involved closing ~12,000 outdated mines during 2011−2019 to improve safety standard, resource utilization, and reduce overcapacity.[5,47] In developed regions such as Beijing, Shanghai, and Zhejiang, the waste sector has become the dominant source (Figure 4). Despite recent policy interventions lead to a measurable decline,[1] the emissions and proportional contribution remain substantial (Figure 5c). Spatial polarization further demonstrates the influence of developmental pathways on methane emissions in different regions during their economic transition.

3.3. Environmental Policies and Investments Emerge as Key Factors to Provincial Disparities

Provincial disparities in methane emissions exhibited a significant increasing trend from 1949 to 2022 ($p < 0.001$), while the trajectory varied across historical periods (Figure 6a).

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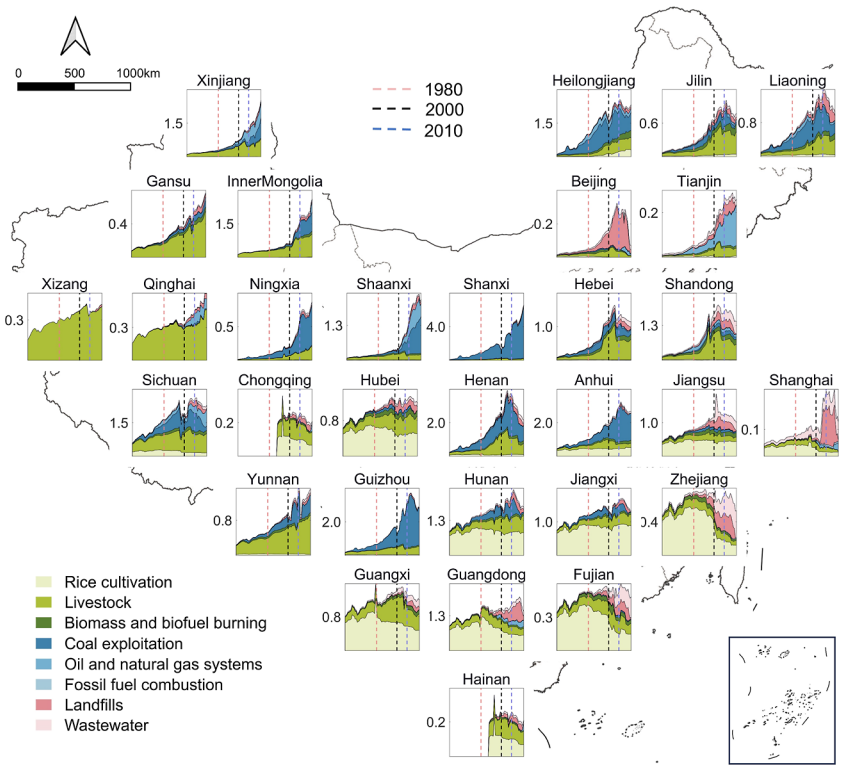


Figure 4. Historical changes of anthropogenic methane emissions in Mainland China across provinces from 1949 to 2022.

Before 1978, interprovincial differences were dominated by the agricultural sectors, accounting for 90.7% of national variation. However, the overall disparity grew marginally during this phase, with a modest decadal increase of 0.04 Tg^2. Between 1978 and 2000, provincial disparities intensified significantly. The contribution of agriculture to these disparities declined from 73.2% to 50.7%, while that of the energy sectors grew steadily, reaching 25.2% by 2000. The year 2000 is a turning point that the later period witnessed an accelerated divergence, with disparities surging at 0.7 Tg^2 per decade. The disparities of energy sectors became the dominant source, averaging 73.6% of the total, almost double the share during 1978−2000, and exceeding 90% in the last five years. Notably, agricultural-sector variation remained stable, increasing by only 13% from 1978 to 2022, far below the growth in the energy (33-fold) and waste sectors (76-fold). This persistent stability in agricultural sectors reflects nationwide commitments to food security, which maintains a baseline level of production across provinces. By contrast, the rapid growth in energy sectors underscores considerable imbalances in regional development pace, resource endowments, and economic trajectories.

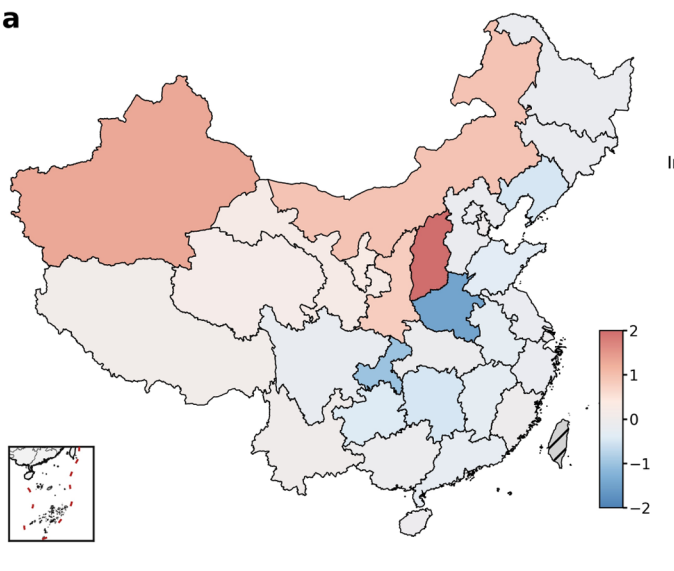
The historical dynamics of provincial disparities were initially dominated by southern agriculture and later by the western energy sectors (Figure 6b−e). Contributions from the waste sectors also increased by comparison to the earlier period (Figure 6e). Energy-sector contributions reveal a clear divergence between northwest and southwest regions as resource-dependent provinces like Shanxi, Xinjiang, and Guizhou disproportionately dominate provincial disparities during 2000−2022. From a sectoral perspective, rice cultivation and coal mining have been the primary factors behind the enlarged disparities (Figure 6b−e). In 1978, rice cultivation was the dominant contributor, led by Guangdong (19.7%), Hunan (14.3%), and Jiangxi (10.8%). Coal mining also played a notable role in provinces such as Shanxi (17.3%), Heilongjiang (4.8%), and Henan (3.6%). By 2022, coal mining had become the main source, dominated by Shanxi (62.9%), Guizhou (17.5%), and Anhui (10.9%).

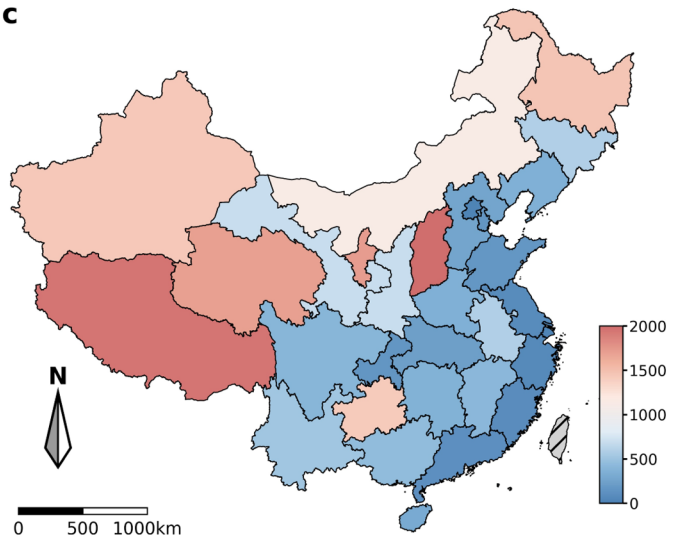
China's economic transition is associated with rising disparities of methane emissions across provinces after 1995, which involves structural shifts from agriculture to industry and technology, together with evolving environmental policies (Figure 7). The period 2010−2012 marks a critical turning point, coinciding with the growing influence of environmental policies related to carbon emissions. Industrial and agricultural development constituted primary factors of rising disparities prior to 2010. Provincial disparities in energy and industrial structures consistently contributed 28.2% to overall disparities, predominantly manifested through uneven energy industry investment and energy consumption (Figure 7 and Table S1). Agricultural activity contributed 17.3% on average during 1995−2009, but its relative share declined after 2010 to a mean of 8.2%, consistent with shifting factors toward industry and policy-related factors. Additionally, consequent regional gaps in sewage and landfill treatment capacity accounted for 13.4% of disparities on average before 2005.

The impact of environmental policy enforcement gradually aligns with that of economic factors after 2010, with its contribution averaging 45.1% (Figure 7). It suggests that the rising disparities of methane emissions across provinces stems from dual forces. First, the uneven implementation of China's 11th and later Five-Year Plan environmental policies on provinces may create methane control gaps, from the aspect of comprehensiveness and depth.[39] This disparity is exemplified by the stark regional imbalance in environmental investment (69% in eastern provinces vs 31% in the west during 2010−2022), where annual methane special policy count and annual methane special policy score, alongside the original cumulative

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b

	TAE	RC	LS	BB	CE	ONG	FF	LF	WW
Beijing	-0.28	0.00	-0.04	-0.01	-0.00	0.01	0.00	-0.25	0.00
Tianjin	-0.03	0.01	-0.02	0.00	0.00	0.07	0.00	-0.12	0.02
Hebei	-0.07	-0.00	0.01	0.01	-0.06	-0.00	0.00	-0.04	0.02
Shanxi	2.43	0.00	0.08	0.01	2.21	0.14	0.00	-0.02	0.01
InnerMongolia	1.15	0.00	0.21	0.07	0.38	0.48	0.00	-0.03	0.04
Liaoning	-0.28	-0.01	-0.03	0.03	-0.22	0.00	0.00	-0.07	0.02
Jilin	-0.01	0.02	0.06	0.07	-0.17	0.01	0.00	-0.03	0.03
Heilongjiang	0.30	0.10	0.11	0.14	-0.10	0.03	0.00	-0.01	0.03
Shanghai	-0.16	-0.00	-0.02	-0.00	0.00	0.03	0.00	-0.17	0.01
Jiangsu	-0.09	-0.01	-0.01	0.00	-0.01	-0.00	0.00	-0.08	0.02
Zhejiang	-0.14	-0.04	-0.02	-0.00	-0.00	0.00	0.00	-0.10	0.03
Anhui	-0.17	0.02	-0.01	0.02	-0.17	0.00	0.00	-0.04	0.02
Fujian	-0.07	-0.01	-0.02	-0.00	-0.01	0.00	0.00	-0.07	0.04
Jiangxi	-0.18	-0.03	0.02	0.00	-0.13	0.00	0.00	-0.05	0.01
Shandong	-0.13	-0.00	-0.06	0.01	-0.02	-0.01	0.00	-0.09	0.04
Henan	-0.47	-0.00	-0.10	0.01	-0.35	-0.00	0.00	-0.04	0.03
Hubei	-0.06	0.01	-0.00	0.00	-0.03	-0.00	0.00	-0.05	0.01
Hunan	-0.21	-0.01	0.04	0.00	-0.21	0.00	0.00	-0.04	0.00
Guangdong	-0.14	-0.04	-0.02	-0.00	0.00	0.01	0.00	-0.09	0.00
Guangxi	-0.11	-0.08	-0.02	-0.00	-0.00	0.00	0.00	-0.03	0.02
Hainan	-0.26	-0.14	-0.14	-0.01	0.00	0.11	0.00	-0.08	0.00
Chongqing	-1.02	-0.01	-0.00	-0.01	-1.04	0.10	0.00	-0.07	0.00
Sichuan	-0.10	-0.01	0.02	0.00	-0.22	0.14	0.00	-0.05	0.02
Guizhou	-0.55	-0.01	0.05	-0.01	-0.56	0.01	0.00	-0.04	0.00
Yunnan	-0.01	-0.01	0.09	0.01	-0.12	-0.00	0.00	-0.01	0.02
Xizang	-0.14	-0.00	-0.25	-0.00	0.00	0.00	0.00	0.10	0.00
Shaanxi	0.51	-0.00	-0.00	0.00	0.47	0.05	0.00	-0.02	0.01
Gansu	0.26	-0.00	0.19	0.02	0.01	0.05	0.00	-0.01	0.00
Qinghai	0.64	0.00	0.57	-0.00	-0.03	0.01	0.00	0.10	0.00
Ningxia	0.63	-0.02	0.36	0.00	0.40	0.02	0.01	-0.12	-0.01
Xinjiang	1.16	-0.00	0.27	0.01	0.72	0.17	0.00	-0.02	0.00

Figure 5. Spatial differences of the changes in methane emissions between 2010 and 2022 across provinces (unit: Tg) (a). Changes of methane emissions between 2010 and 2022 across sectors and provinces (b) (unit: Tg). The abbreviations are total anthropogenic emissions (TAE), rice cultivation (RC), livestock (LS), biomass and biofuel burning (BB), coal exploitation (CE), oil and natural gas systems (ONG), fossil fuel combustion (FF), landfills (LF), and wastewater (WW). Ratio from methane emissions to GDP at 2022 (c) (unit: tCH_4 per billion).

stocks are key contributors to rising disparities (Table S1). In other words, provinces that accumulate denser and higher-scoring methane-related measures over time, and that continue to add new measures in later years, are more strongly differentiated from provinces with thinner policy trajectories. Second, these governance asymmetries interact with concurrent industrial relocation of carbon-intensive production toward central and western provinces. As eastern regions tighten environmental controls and upgrade industrial structure, emission-intensive capacity may concentrate westward, reinforcing an intergenerational emissions gap relative to cleaner industries in the east. Together, uneven policy depth and shifting industrial geography provide a plausible dual explanation for why provincial methane disparities continued to widen even as national environmental governance intensified after 2010.

3.4. Recent Decoupling Masks the Disproportion between Emission and Economic Gains

Economic development has differentially influenced the provincial disparities of methane emissions. The year 2010 marked a transitional point with a shift toward strong decoupling between GDP and methane emissions, though with notable interregional heterogeneity (Figures 8a and S5). Northwest provinces stand as the only area where the growth of methane emissions occurred with rising GDP. They experience a 62.7% increase in methane emissions alongside a 1.8-fold rise in GDP, while southwest provinces achieved a 2.8-fold economic gain coupled with a 22.1% methane reduction (Figure 8b). This divergence suggests a persistent carbon lock-in effect in the northwest, while the southwest exemplifies successful transitional trajectory. In the initial decades after economic reforms, economic growth remained partially tied to traditional industrial and agricultural practices,

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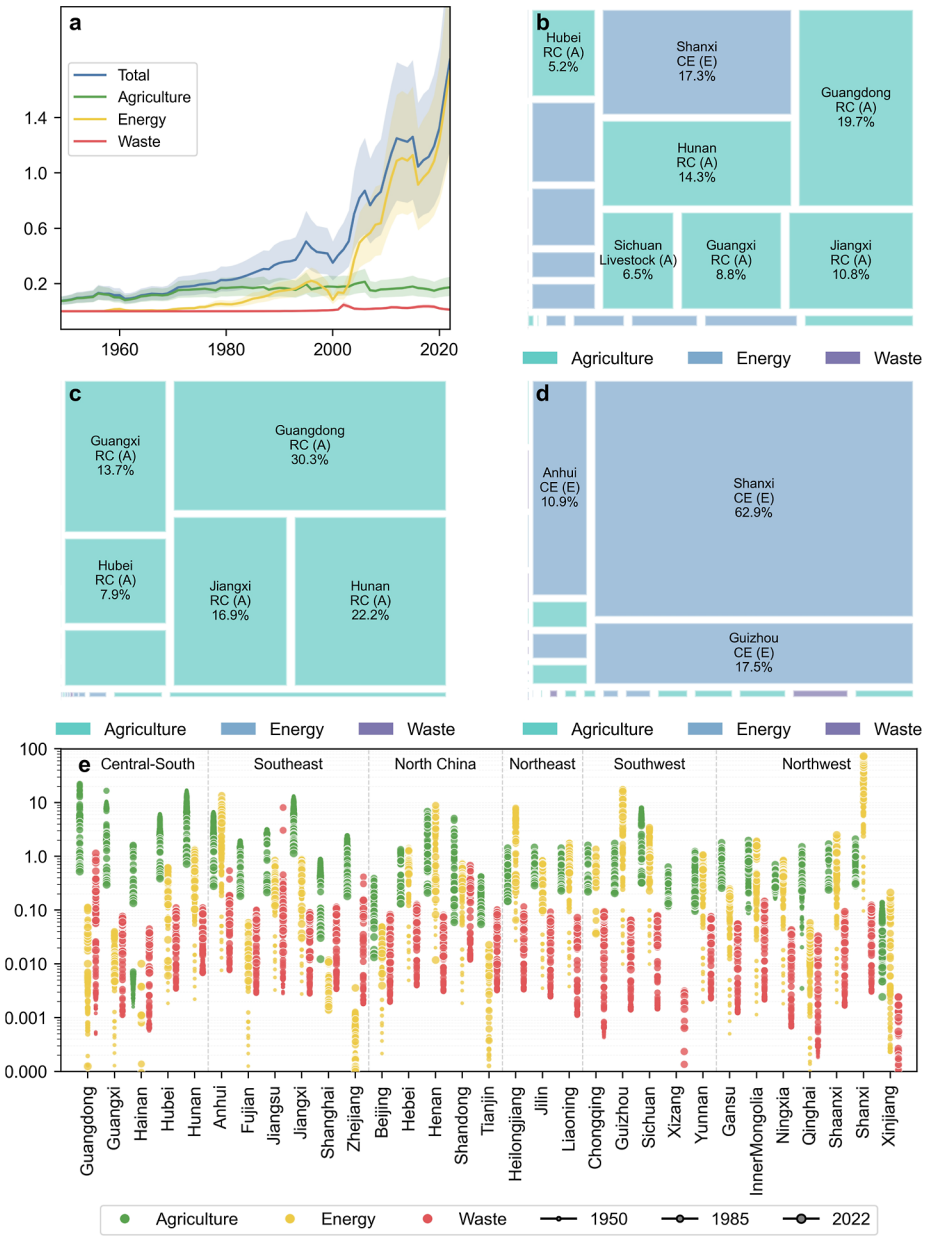


Figure 6. Temporal variations in provincial disparities of methane emissions from agriculture, energy, waste, and the whole sectors (a) (unit: Tg^2). (b–d) Province-sector ranking as their contribution to provincial disparities in 1950, 1978, and 2022, respectively. The abbreviations in panels (b–d) are as follows: RC and CE. (e) Provincial-level composition of provincial disparities: agriculture, energy, and waste sectors, and circle size indicates proximity to 2022 (larger = closer).

yielding a pattern of weak decoupling with decoupling index between 0 and 0.8 (Figure 8a). This is particularly evident in less developed provinces of the southwest, northwest, and northeast region (Figure 8c). The regional decoupling post-2010 is largely attributable to differential performance of the energy sector (Figure S6). Eastern regions attain strong decoupling in energy-related emissions, despite some fluctuations. Conversely, the agricultural sector demonstrated minimal decoupling, with a slight increase (Figure S6), likely reflecting the constraints imposed by national food security strategies.

Northern and southern regions display an asymmetrical trend of the disproportion between GDP and methane emissions (Figure 8d). This north–south divide suggests that northern regions face higher emission levels and potentially greater mitigation challenges, yet receive lower economic returns. Specially, economic growth in northern regions has corresponded to a disproportionately faster rise in methane emissions (Figure 8d). This is evident in the northwest, where each unit of GDP was associated with two units of methane emissions over the recent cycle. In contrast, southern regions show a declining trend of the disproportion, with the southeast showing the largest improvement (Figure 8d). The share of methane emissions decreases but economic gains increase. A similar pattern is observed in the agricultural sector, while the energy sector plays a dominant role in affecting regional disparities due to its larger absolute emissions and spatial heterogeneity (Figure S6).

Less developed western and northeastern regions have maintained a long-term disproportion since 1980, character-

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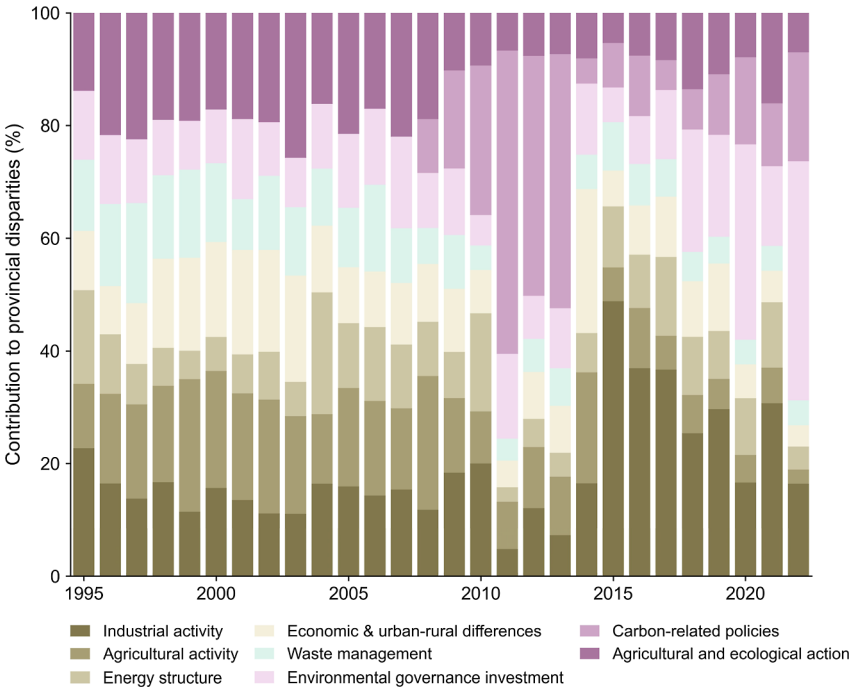


Figure 7. Explanation of different indicators to provincial disparities of methane emissions from 1995 to 2022. Brown shades represent economic activity indicators (i.e., industrial activity, agriculture activity, energy structure, and urban-rural economic differences), purple represents environmental policy and investment indicators (i.e., environmental governance investments, carbon-related policies, and agriculture or ecological actions), and green represents waste management indicators.

ized by high methane emissions without commensurate economic gains (Figure 8d). Northern China and the southeastern coast, as affluent regions, exhibit lower emissions alongside higher economic output. Notably, within both northern and southern zones, less developed western areas show more acute imbalances than their eastern counterparts. This indicates that eastern provinces transition toward technology- and capital-intensive industries (e.g., 17.22−68.81 tCH_4/billion CNY in Beijing, Shanghai, and Zhejiang). Conversely, western resource-dependent provinces become increasingly specialized in energy processing and other high-emission activities (e.g., 1416.04−2579.73 tCH_4/billion CNY in Guizhou, Ningxia, and Shanxi). This functional divergence entrenches an "emissions westward, benefits eastward" paradigm. As of 2022, the northwest had the largest imbalance, which contributed the highest methane emissions but the lowest economic benefits (Figure 8e).

4. DISCUSSIONS

Our subnational methane emission inventory (1949−2022) captures the historical evolution of China's methane emissions across divergent development phases, while reflecting how policy interventions and resource distribution impact disparities. The nation's transformation from agriculture-dominant to energy-dominant methane profiles has intensified obviously and caused enlarged provincial disparities. Before reform, disparities largely mirrored agro-ecological and cropping heterogeneity that rice belts in the south generated differentiated but relatively stable spatial contrasts. After reform, the disparities became increasingly tied to extractive energy systems, especially after 2000. The relative stability of agricultural disparities is equally informative. It reflects the homogenizing effect of national food-security governance, which suppresses large province-to-province swings in baseline production. Interpreting western provinces as trapped in high-emission, low-return pathways is therefore a hypothesis suggested by the co-occurrence of rising energy disparities, weak decoupling in the northwest, and persistent emissions−GDP disproportion. Overall, the asymmetric economic development across regions and inconsistent environmental policy implementation are the key explanatory factors of these disparities. The findings are consistent with recent studies,[1,3,28,48] which highlight the pivotal roles of economic growth and policies in shaping methane emissions. Our research extends the analysis to encompass broader historical context.

Our inventory shows that national anthropogenic methane emissions have entered a postpeak decline, with a clear slowdown in growth and an emerging decoupling between economic activity and emission sources, particularly in the energy sectors. This suggests that China is exploring a development pathway designed to harmonize sustained economic growth with a low-carbon transition, attributable to the widespread implementation of energy policies and environmental regulations.[1] The subsequent decline in emissions and the decoupling between economic activity and emission sources, especially the energy sectors, also reflected this point, as consistent with the recent findings.[49] Waste sectors from coastal provinces show a significant decline in recent years (Figure 4). This downward trend accelerated after the COVID-19 pandemic due to mitigation efforts under national and provincial waste management regulations.[39,50]

Provincial disparities increasingly reflect different sectoral pathways: stable agricultural contrasts under food-security governance versus rising energy-related concentration in resource provinces. The results are consistent with a spatial mismatch in which western resource provinces host high emissions without commensurate local gains,[5,51,52] however,

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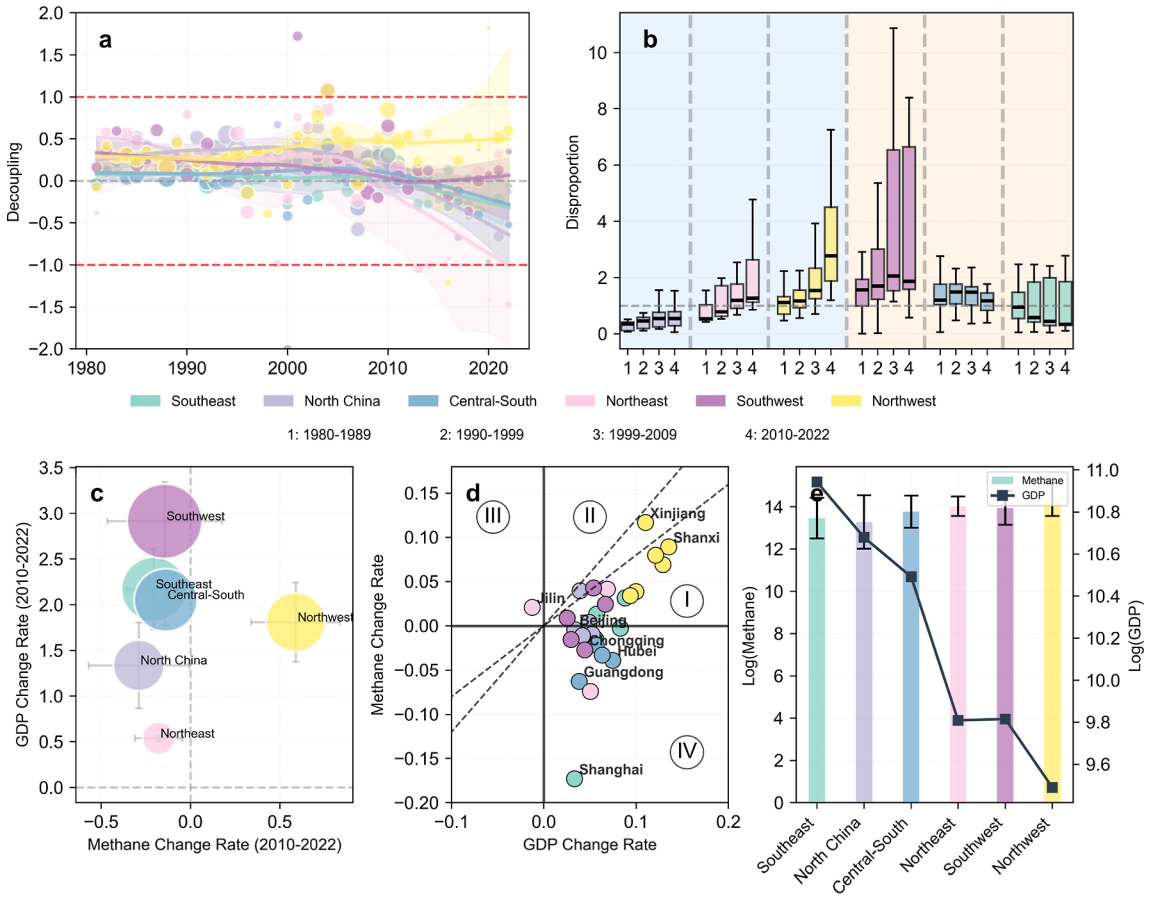


Figure 8. Region decoupling status and the disproportion between methane emissions and GDP since 1980. (a) Decoupling of methane emissions from GDP by region and year. Each dot means the median of all provinces belonging to the same region for specific year. (b) Changes in methane emissions relative to GDP across regions, 2010−2022. (c) Decoupling status by province and region in 2022. Solid dots represent individual provinces. Ellipses indicate regional distributions (constructed with twice the standard error). Roman numerals I−IV denote weak decoupling, expansive negative decoupling, strong negative decoupling, and strong decoupling, respectively. (d) Spatiotemporal variations in the methane−GDP disproportion across all sectors. (e) Regional disparities in methane emissions and GDP in 2022.

methane intensity per unit of provincial GDP alone cannot well establish who benefits from methane-intensive activities or who bears the costs. Resolving that question requires embodied methane flows or consumption-based accounting with accurate provincial input−output linkages. Our results suggest that methane governance may need to look beyond where emissions occur toward why they occur, but this remains a hypothesis rather than a causal claim.

The novelty of this work lies in its extended temporal coverage, which overcomes the limitations of short-window remote-sensing inversions and inventories. This long-term perspective makes it possible to study the decoupling of emissions and economic development across institutional eras. These findings provide not only basis for decarbonization under China's carbon neutrality framework but also guidance to avoid replicating high-emission pathways for the Global South countries. Additionally, province−sector attribution grounds disparities in specific emission systems, tracing shifts from southern rice cultivation to western coal exploitation. This thus may bridge broad spatial patterns with actionable mitigation-relevant interventions. The future work will model methane trajectories and quantify the trade-offs under key policy scenarios, such as energy mix transitions, cropland management innovations, and regional development rebalancing.[53,54] This will translate identified disparities and path dependencies into actionable decarbonization pathways, providing policymakers with the quantitative evidence to design equitable and context-specific mitigation strategies.[2,55]

We identified a turning point of China's anthropogenic methane emissions around 2012, yet this finding remains subject to substantial uncertainty. While it aligns with findings from PKU-CH4_v2,[19] several independent studies indicate that China's methane emissions may still be increasing.[1,22] Early period provincial activity series rely heavily on interpolation, extrapolation, and national-to-provincial allocation, so gap-filling errors and alternative allocation assumptions can still shift both the national trajectory and spatial disparity metrics, even though recent years are better constrained by direct statistics. Our activity-focused sensitivity analyses leave the post-2012 decline, but they also show that quantitative details, including the peak year under some gap-fill extremes and the magnitude of recent change, are not uniquely determined. Emission-factor uncertainty further dominates the combined national envelope relative to activity uncertainty, implying that absolute emission levels and cross-inventory contrasts remain sensitive to methodological choices beyond activity reconstruction alone. In particular, reliance on fixed EF continues to introduce large sectoral uncertainty,[22] especially for rice cultivation, where flooding regimes and human management practices are poorly captured by constant

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Environ. Sci. Technol. XXXX, XXX, XXX−XXX

factors.[56−58] These gaps underscore the need for subsequent top-down satellite inversion to constrain and correct inventory-based estimates.[1]

■ ASSOCIATED CONTENT

Data Availability Statement

China's subnational methane emissions inventory (1949−2022) is publicly available at 10.5281/zenodo.20845351, including codes, sectoral methane emissions, validation data sets, and provincial factors to the disparities of methane emissions. Additional data are provided in the manuscript and Supporting Information Appendix.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.est.6c05777.

Additional methods and analysis, including activity-data availability, comparisons of disparity metrics, semivariance bin-width sensitivity, factor-attribution workflow, regional decoupling and methane−GDP disproportion patterns, selected drivers, provincial rice paddy areas, EF, residue-burning fractions, and waste parameters (PDF)

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Author Contributions

Y.Z. conceptualized the study. F.G. developed and implemented the computer code, performed the analysis, validation, visualized the results, and wrote the manuscript. F.G., F.D., K.F., S.L., P.G., and Y.Z. edited and revised the manuscript.

Notes

The authors declare no competing financial interest.

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